

Body Fat and Weight Loss Calculator Report - MJ Prime

Report Date: 07/07/2025

1. Current Profile

Height:	5'9" (175.26 cm)
Current Weight:	266.80 lbs
Current Body Fat:	36.00%
Current Lean Mass:	170.75 lbs
Current Fat Mass:	96.05 lbs
Age:	46 years
Gender:	Male

2. Target Goals

Goal Weight:	217.00 lbs
Goal Body Fat:	9.90%
Timeline:	07/07/25 to 10/13/25
Total Duration:	14 weeks

3. Current Activity Profile

Activity Level:	Lightly Active (exercise 1-3 days/week)
Training Experience:	Advanced
Workout Type:	Bodybuilding
Weekly Workout Frequency:	3 days
Currently Using Resistance Training:	Yes
Athlete Status:	Yes

Understanding Your Activity Profile:

- **Activity Level:** Your Lightly Active (exercise 1-3 days/week) lifestyle contributes significantly to your daily caloric expenditure.
- **Training Experience:** As a Advanced trainee, this affects your potential muscle gain and recovery needs.
- **Workout Frequency:** Training 3 days per week helps optimize your metabolic rate and recovery periods.
- **Resistance Training:** Your use of resistance training enhances muscle preservation during weight loss.

4. Current Metabolic Profile

Base Metabolic Rate (RMR):	2283.00 cal/day
Total Daily Energy Expenditure (TDEE):	4395.00 cal/day
Thermic Effect of Food (TEF):	352.45 cal/day
Non-Exercise Activity Thermogenesis (NEAT):	200.00 cal/day
Recommended Daily Calorie Intake:	2363.72 cal/day

Understanding Your Metabolic Profile:

- **RMR (Resting Metabolic Rate):** Your body burns 2283.00 calories daily at rest. This is calculated using the Mifflin-St Jeor equation, adjusted for your body composition.
- **TDEE (Total Daily Energy Expenditure):** Based on your activity level, you burn approximately 4395.00 calories per day.
- **TEF (Thermic Effect of Food):** You burn about 352.45 calories digesting food, particularly protein.
- **NEAT:** Your daily activities burn approximately 200.00 additional calories.

Additional Lifestyle Factors

Target Daily Protein Intake:	265.0 grams
Diet Type:	keto
Reported PED Use:	Yes
Reported Sleep Quality:	poor

Impact of These Factors:

- **Protein Intake (265.0g):** Adequate protein is crucial for muscle repair, growth, and satiety. Higher protein intake also increases TEF.
- **Diet Type (keto):** Your chosen diet type influences macronutrient distribution, which can affect energy levels, satiety, and adherence. Specific diet types also have varying impacts on TEF and nutrient partitioning.
- **PED Use (Yes):** If PEDs are used, this can significantly alter metabolic rate, muscle protein synthesis, and nutrient partitioning, leading to faster changes than naturally achievable. Calculations are adjusted accordingly.
- **Sleep Quality (poor):** Sleep is vital for recovery, hormone regulation (including those related to appetite and muscle growth), and overall well-being. Poor sleep can hinder progress.

5. Workout Analysis

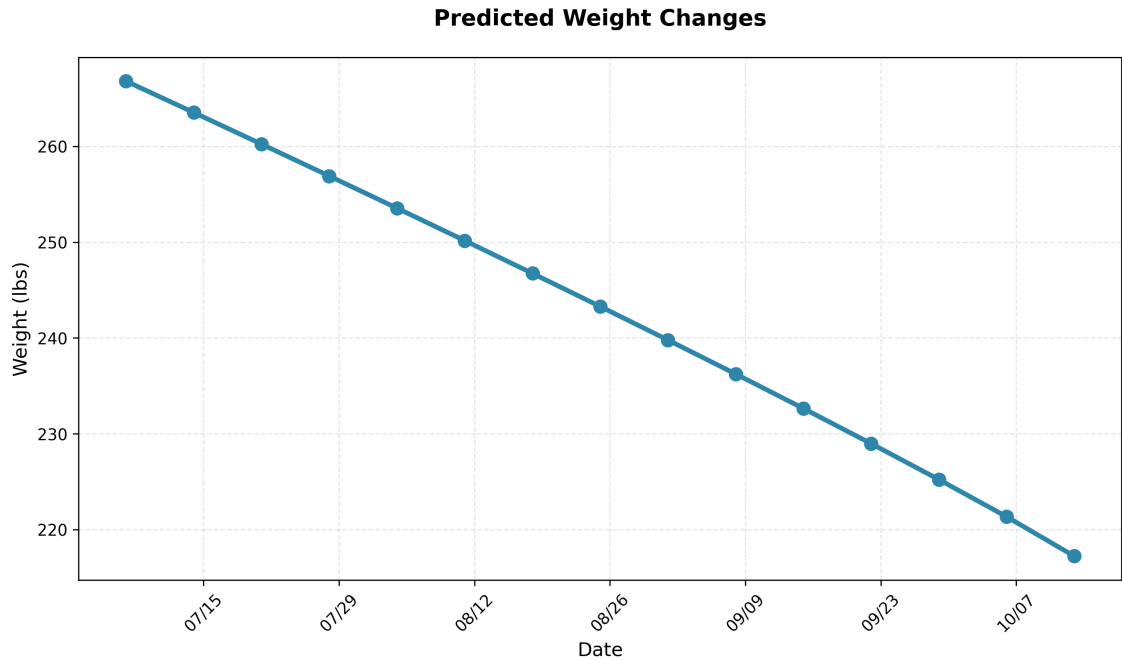
Type:	Bodybuilding
Frequency:	3 days/week
Volume Score:	0.47 - Moderate
Intensity Score:	0.88 - High
Frequency Score:	1.00 - High

Impact on Your Caloric Needs:

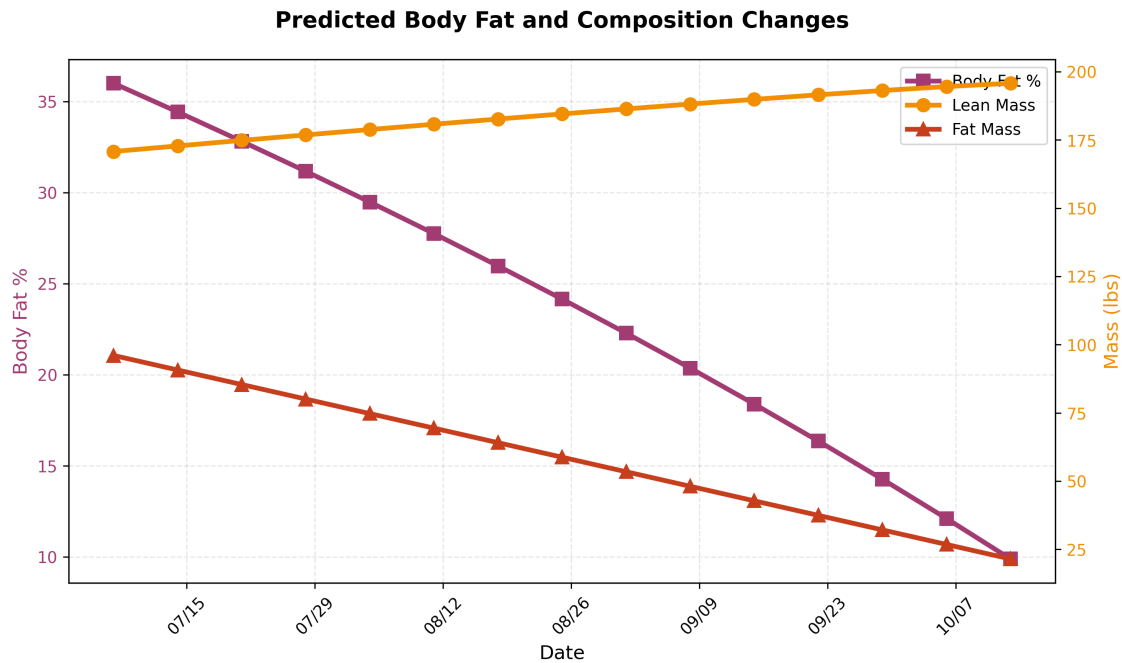
- **Volume Score 0.47:** Your training volume contributes approximately 235.71 additional calories to your daily expenditure.
- **Intensity Score 0.88:** Your training intensity increases your metabolic rate by about 13.20% for up to 48 hours post-workout.
- **Frequency Score 1.00:** Training 3 days per week optimizes your caloric burn and recovery periods.

6. Predicted Progress Charts

Predicted Weight Changes



Predicted Body Fat Changes



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7. Weekly Progress Forecast

Date	Weight (lbs)	Body Fat %	Daily Cal Intake	TDEE	Weekly Cal Output	Total Weight Lost (lbs)	Lean Mass (lbs)	Fat Mass (lbs)	Muscle Gain (lbs)	RMR
07/07/25	266.80	36.00	2363.72	4395.00	0.00	0.00	170.75	96.05	0.00	2283.00
07/14/25	263.52	34.43	2363.72	4395.00	14218.93	3.28	172.80	90.72	0.28	2283.00
07/21/25	260.23	32.82	2330.98	4370.00	14273.11	6.57	174.83	85.40	0.28	2267.00
07/28/25	256.90	31.17	2296.65	4344.00	14331.47	9.90	176.83	80.07	0.28	2250.00
08/04/25	253.55	29.48	2262.66	4319.00	14394.35	13.25	178.80	74.74	0.27	2234.00
08/11/25	250.16	27.75	2226.91	4293.00	14462.63	16.64	180.74	69.42	0.27	2217.00
08/18/25	246.74	25.98	2189.21	4266.00	14537.50	20.06	182.65	64.09	0.27	2200.00
08/25/25	243.29	24.15	2151.34	4240.00	14620.61	23.51	184.52	58.77	0.26	2183.00
09/01/25	239.79	22.29	2110.96	4213.00	14714.30	27.01	186.35	53.44	0.26	2166.00
09/08/25	236.25	20.37	2068.56	4186.00	14822.07	30.55	188.13	48.11	0.25	2148.00
09/15/25	232.65	18.39	2023.35	4159.00	14949.53	34.15	189.86	42.79	0.25	2131.00
09/22/25	228.98	16.36	1972.93	4131.00	15106.51	37.82	191.52	37.46	0.25	2113.00
09/29/25	225.23	14.27	1915.48	4103.00	15312.62	41.57	193.10	32.14	0.24	2094.00
10/06/25	221.36	12.11	1843.01	4074.00	15616.92	45.44	194.55	26.81	0.24	2076.00
10/13/25	217.24	9.89	1727.49	4044.00	16215.54	49.56	195.75	21.48	0.24	2056.00

8. Body Composition Changes

Body Fat Category	Body Fat %	Date Reached	Description	Est. Time to Six-Pack
Obese	36.00%	07/07/25	Significant excess fat all around	12+ months
Obese	34.43%	07/14/25	Significant excess fat all around	12+ months
Obese	32.82%	07/21/25	Significant excess fat all around	12+ months
Obese	31.17%	07/28/25	Significant excess fat all around	12+ months
High Body Fat	29.48%	08/04/25	Excess fat all around, round physique	6-12 months
High Body Fat	27.75%	08/11/25	Excess fat all around, round physique	6-12 months
High Body Fat	25.98%	08/18/25	Excess fat all around, round physique	6-12 months
Above Average	24.15%	08/25/25	No visible abs, excess fat	4-6 months
Above Average	22.29%	09/01/25	No visible abs, excess fat	4-6 months
Above Average	20.37%	09/08/25	No visible abs, excess fat	4-6 months
Average	18.39%	09/15/25	Little muscle definition, soft look	3-4 months
Average	16.36%	09/22/25	Little muscle definition, soft look	3-4 months
Fit	14.27%	09/29/25	Some muscle definition, minimal fat	2-3 months
Fit	12.11%	10/06/25	Some muscle definition, minimal fat	2-3 months
Very Lean	9.89%	10/13/25	Excellent definition, very low fat	Already there

9. Expected Results

Total Weight Loss:	49.56 lbs
Fat Loss:	26.11%
Muscle Gain:	25.00 lbs

Average Weekly Changes:

Weight Loss:	3.54 lbs
Muscle Gain:	1.79 lbs

Understanding Your Results:

- **Total Weight Loss:** Your projected loss of 49.56 lbs is designed to be sustainable and preserve muscle mass.
- **Fat Loss:** A reduction of 26.11% body fat represents a healthy rate of fat loss.
- **Muscle Changes:** The program aims to gain/maintain 25.00 lbs of muscle while losing fat.
- **Weekly Progress:** Targeting 3.54 lbs per week ensures steady, sustainable progress.

10. Metabolic Adaptation Forecast

Initial Week:	1.00 multiplier
Final Week:	0.90 multiplier
Expected Adaptation:	9.94% change
Lean Mass Preservation:	114.64%

Understanding Metabolic Adaptation:

- **Initial Response:** Your metabolism starts at 1.00x baseline, reflecting your current activity level.
- **Progressive Changes:** A 9.94% adaptation is expected as your body becomes more efficient.
- **Muscle Preservation:** The program is designed to maintain 114.64% of your lean mass.

11. Final Phase Targets

Maintenance Calories:	4044.00 calories/day
Weekly Energy Output:	16215.54 calories

Understanding Your Final Targets:

- **Maintenance Level:** Your final daily maintenance of 4044.00 calories reflects your new metabolic baseline.
- **Weekly Output:** Maintaining a weekly energy output of 16215.54 calories will help sustain your results.
- **Long-term Success:** These targets are designed to help you maintain your new weight and body composition.

12. Input Parameter Analysis

All Input Parameters Used in Calculations:

This section confirms that all provided input parameters are directly used in the body composition calculations.

Parameter	Value	Used in Calculation	Calculation Component
Activity Level	2	✓ Yes	TDEE calculation (activity multiplier 1.2-1.9)
Athlete Status	True	✓ Yes	RMR boost (10% increase)
Bodybuilder Mode	True	✓ Yes	Calorie limits and muscle gain adjustments
Current Body Fat	36.0	✓ Yes	Body composition calculations
Current Weight	266.8	✓ Yes	RMR calculation (Mifflin-St Jeor), TDEE, muscle gain estimation
Date of Birth	070478	✓ Yes	Age calculation for RMR and muscle gain adjustments
Diet Type	keto	✓ Yes	Muscle gain and fat loss multipliers
Exercise Type	resistance	✓ Yes	NEAT boost and muscle gain adjustments
Gender	Male	✓ Yes	RMR calculation (different formulas for male/female)
Goal Body Fat	9.9	✓ Yes	Dual-goal deficit calculations
Goal Weight	217.0	✓ Yes	Dual-goal deficit calculations
Height (Feet)	5	✓ Yes	RMR calculation (converted to height_cm)
Height (Inches)	9	✓ Yes	RMR calculation (converted to height_cm)
Job Activity	1	✓ Yes	NEAT estimation (100-700 calories)
Leisure Activity	1	✓ Yes	NEAT estimation (50-350 calories)
PED Use	True	✓ Yes	Muscle gain multiplier (1.5x) and fat loss ratio
Protein Intake	265.0	✓ Yes	TEF calculation (25% thermic effect) and muscle gain
Resistance Training	True	✓ Yes	Muscle gain calculation (0.05 baseline if False, full calculation if True)
Sleep Quality	poor	✓ Yes	Muscle gain multiplier (1.0 vs 0.8)
Training Experience	Advanced	✓ Yes	Base muscle gain rates (0.0025-0.0125 weekly)

Input Reliability Analysis:

Category	Score	Assessment
Anthropometric Data	100.0/100	Excellent
Activity Data	100.0/100	Excellent
Goal Data	80.0/100	Good
Data Completeness	100.0/100	Complete

13. AI Confidence Analysis

AI-Powered Assessment of Your Plan:

This analysis uses advanced AI to evaluate the reliability and feasibility of your body composition goals.

Analysis Component	Score	Assessment
Overall Confidence	65.0/100	Moderate
Input Reliability	95.0/100	Very Reliable
Calculation Accuracy	90.0/100	Very Accurate
Goal Feasibility	60.0/100	Challenging

Score Explanations:

Component	Score	AI Explanation
Overall Confidence	65.0/100	The transformation plan shows moderate feasibility but faces significant challenges. The target weekly weight loss of 3.3 lbs is aggressive and may be unsustainable long-term. The transition from 36% to 9.9% body fat represents a major physiological change that requires careful planning and execution.
Input Reliability	95.0/100	Input data shows high reliability with perfect scores for anthropometric, activity data, and completeness. The slightly lower goal data reliability (80/100) slightly impacts overall input reliability, but the core measurements appear accurate and well-documented.
Calculation Accuracy	90.0/100	The TDEE calculation and caloric deficit recommendations follow established principles. The 2363 calorie target represents a 2031 calorie deficit, which aligns with the projected weight loss. However, the aggressive deficit may need adjustment as body composition changes.
Goal Feasibility	60.0/100	The goal of reaching 9.9% body fat from 36% is ambitious. While possible, this represents a loss of approximately 49.8 lbs of pure fat mass while maintaining lean mass, which requires precise nutrition and training protocols. The absence of a defined timeframe makes feasibility assessment challenging.

⚠ Important Warnings:

- Weekly weight loss target exceeds recommended safe range of 1-2 lbs
- Significant caloric deficit may impact adherence and metabolism
- Reaching 9.9% body fat requires exceptional dedication and may not be sustainable
- No specified timeframe could lead to unrealistic expectations
- Risk of lean mass loss with aggressive deficit

AI Recommendations:

- Consider reducing weekly weight loss target to 1.5-2 lbs for sustainability
- Implement a phased approach with intermediate body fat percentage goals
- Add specific timeframe to better structure the transformation plan
- Include regular diet breaks to manage metabolic adaptation
- Incorporate strength training to preserve lean mass

- Consider setting a more moderate body fat goal initially (15-18%)

Detailed Confidence Factors:

Factor	Score
Timeframe Realism	0.0/100
Calorie Deficit Sustainability	70.0/100
Body Composition Feasibility	65.0/100
Data Quality	95.0/100

Detailed AI Analysis Summary:

This transformation plan presents a comprehensive but aggressive approach to body recomposition. The high-quality input data provides a solid foundation for planning, but several key factors require attention. The absence of a defined timeframe significantly impacts the ability to assess realistic progression. The calculated daily deficit of 2031 calories, while mathematically sound for the target weight loss, may be too aggressive for long-term adherence and metabolic health. The transition from 36% to 9.9% body fat represents a loss of approximately 49.8 lbs of fat mass while ideally maintaining lean mass - a challenging goal requiring perfect execution of diet and training. The plan would benefit from a more moderate initial approach, potentially targeting a higher initial body fat percentage and incorporating planned diet breaks. The high data quality and accurate calculations provide a good framework, but the aggressive nature of the goals and deficit suggest a need for modification to improve long-term success probability. Implementation of the suggested modifications would significantly increase the overall confidence score.

14. Comprehensive Parameter Reference Guide

This section details exactly how each input parameter affects your calculations and results:

Parameter Usage in Calculations:

Parameter	Your Value	Usage Location	Impact on Results
Age	46 years	RMR Calculation (Mifflin-St Jeor)	Base metabolic rate decreases ~1-2% per decade after age 30
Weight	266.8 lbs	RMR, TDEE, Body Composition	Primary factor in all metabolic calculations - higher weight = higher energy needs
Height	5'9" (175.26 cm)	RMR Calculation, BMI	Taller individuals have higher metabolic rates due to larger body surface area
Gender	Male	RMR Calculation (different formulas)	Males typically have 10-15% higher RMR than females due to muscle mass differences
Activity Level	Lightly Active (exercise 1-3 days/week)	TDEE Multiplier	Multiplies RMR by 1.2-1.9 based on activity level
Body Fat %	36.0%	Lean Body Mass, Metabolic Calculations	Lower body fat = higher lean mass = higher metabolic rate per pound
Goal Weight	217.0 lbs	Calorie Adjustment, Macro Distribution	Target weight determines caloric deficit/surplus needed
Resistance Training	Yes	PRIME_Diet_Calculations_v2.py:161	Increases protein needs and metabolic boost
Athlete Status	Yes	PRIME_Utills.py:105-106	Adds 10% RMR boost for enhanced metabolic capacity
Workout Type	Bodybuilding	Macro Distribution, TEF Calculations	Bodybuilding training affects protein/carb/fat ratios and thermic effect
Experience Level	Advanced	NEAT Estimation, Performance Adjustments	Advanced affects recovery needs and energy expenditure patterns
PED Use	Yes	PRIME_Calculations.py, PRIME_Diet_Calculations_v2.py	Increases muscle gain by 50% (1.5x multiplier) and improves fat loss ratio by 20%

Calculation Impact Summary:

- **Your RMR:** 2283 calories - calculated using age (46), weight (266.8lbs), height (5'9" (175.26 cm)), and gender (Male)

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- **Your TDEE:** 4395 calories - RMR multiplied by activity factor for Lightly Active (exercise 1-3 days/week) activity level
- **Your Goal Calories:** 2364 calories - TDEE adjusted for your goal
- **Body Fat Impact:** 36.0% body fat indicates 171lbs lean mass, affecting metabolic calculations
- **Athlete Bonus:** 10% RMR increase applied due to athlete status
- **PED Enhancement:** 50% muscle gain increase and 20% improved fat loss ratio applied

Code References:

- **Resistance Training Check:** PRIME_Diet_Calculations_v2.py line 161 - Uses 0.05 baseline if False
- **Athlete RMR Boost:** PRIME_Utills.py lines 105-106 - Adds 10% if True
- **Diet Type Multipliers:** Multiple locations with fat loss/muscle gain adjustments
- **Exercise Type NEAT:** NEAT estimation and performance adjustment calculations
- **Activity Level Multipliers:** Standard TDEE calculation with activity-specific factors
- **PED Use Multipliers:** PRIME_Calculations.py line 81 - 1.5x muscle gain;
PRIME_Diet_Calculations_v2.py lines 109-110, 166-167 - Enhanced fat loss and muscle gain

14. Comprehensive Reference Guide

Activity Levels and Caloric Impact

Level	Description	TDEE Multiplier	Daily Caloric Impact	Example Activities
Sedentary	Little to no exercise, desk job	1.2	+20% above RMR	Office work, driving, watching TV
Lightly Active	Light exercise 1-3 days/week	1.375	+37.5% above RMR	Walking, light gardening, yoga
Moderately Active	Moderate exercise 3-5 days/week	1.55	+55% above RMR	Jogging, cycling, recreational sports
Very Active	Heavy exercise 6-7 days/week	1.725	+72.5% above RMR	Construction work, competitive sports
Extremely Active	Very heavy exercise, physical job	1.9	+90% above RMR	Professional athletes, heavy manual labor

Training Experience and Muscle Gain Potential

Level	Years of Training	Monthly Muscle Gain	Caloric Requirements	Recovery Needs
Beginner	0-2 years	0.5-1% body weight	+500 cal/day surplus	24-36 hours between sessions
Intermediate	2-4 years	0.25-0.5% body weight	+300-400 cal/day surplus	36-48 hours between sessions
Advanced	4+ years	0.125-0.25% body weight	+200-300 cal/day surplus	48-72 hours between sessions

Workout Types and Metabolic Impact

Type	Primary Focus	Caloric Burn	EPOC Effect	Recovery Demand
Bodybuilding	Muscle growth and aesthetics	600-800 cal/session	24-36 hours	High protein needs (1g/lb)
Strength Training	Maximal strength	400-600 cal/session	24-48 hours	High CNS recovery needs
Powerlifting	Specific lift performance	500-700 cal/session	36-48 hours	Very high CNS demands
Olympic Weightlifting	Power and technique	600-800 cal/session	24-36 hours	Technical recovery focus
CrossFit	Overall fitness	800-1000 cal/session	36-48 hours	High systemic demands
Calisthenics	Bodyweight control	400-600 cal/session	24-36 hours	Joint/tendon focus

Resistance Training Impact

Status	Muscle Preservation	Metabolic Impact
With RT	90-100%	+100-300 cal/day
Without RT	60-80%	No additional benefit

Impact on Calculations: Resistance training increases your daily caloric needs by 100-300 calories through increased muscle mass and EPOC (Excess Post-exercise Oxygen Consumption).

Athlete Status Effects

Status	Metabolic Rate	Nutrient Partitioning
Athlete	+10-15% higher	Enhanced efficiency
Non-Athlete	Standard	Normal efficiency

Impact on Calculations: Athletic status can increase your TDEE by 10-15% due to enhanced metabolic efficiency and muscle mass.

Volume Score Components

Score Range	Sets per Muscle	Weekly Volume
0.0-0.2	1-5 sets	Low stimulus
0.2-0.4	6-10 sets	Maintenance
0.4-0.6	10-15 sets	Growth stimulus
0.6-0.8	15-20 sets	Optimal growth
0.8-1.0	20+ sets	Maximum stimulus

Intensity Score Impact

Score Range	% of 1RM	Rep Range
0.0-0.2	30-50%	15+ reps
0.2-0.4	50-65%	12-15 reps
0.4-0.6	65-75%	8-12 reps
0.6-0.8	75-85%	5-8 reps
0.8-1.0	85%+	1-5 reps

Frequency Score Effects

Score Range	Sessions/Week	Recovery Time
0.0-0.2	1x/week	7 days
0.2-0.4	2x/week	3-4 days
0.4-0.6	3x/week	2-3 days
0.6-0.8	4x/week	1-2 days
0.8-1.0	5+x/week	<1 day

Calculation Formulas and Methodology

Component	Formula	Variables	Example
RMR (Male)	$(10 \times \text{weight kg}) + (6.25 \times \text{height cm}) - (5 \times \text{age}) + 5$	Weight, Height, Age	70kg man: 1,745 cal
RMR (Female)	$(10 \times \text{weight kg}) + (6.25 \times \text{height cm}) - (5 \times \text{age}) - 161$	Weight, Height, Age	60kg woman: 1,374 cal
TDEE	RMR $\times$ Activity Multiplier + Exercise Calories	RMR, Activity Level	$1,745 \times 1.55 = 2,705$ cal
Caloric Deficit	TDEE - Daily Intake	TDEE, Target Loss	$2,705 - 2,205 = 500$ cal
Weekly Fat Loss	$(\text{Weekly Deficit}) \div 3,500$	Daily Deficit $\times 7$	$(500 \times 7) \div 3,500 = 1$ lb
Muscle Gain	$(\text{Body Weight} \times \text{Experience Factor})$	Weight, Training Level	$70\text{kg} \times 0.01 = 0.7\text{kg/month}$